

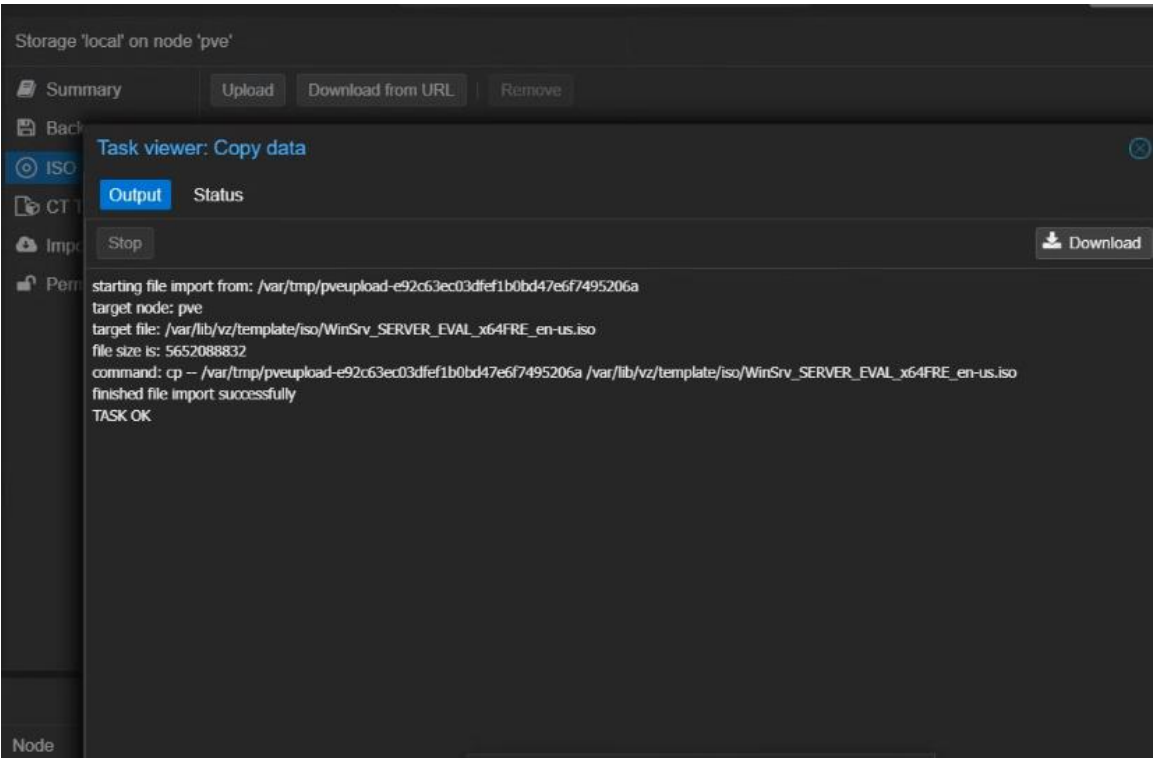
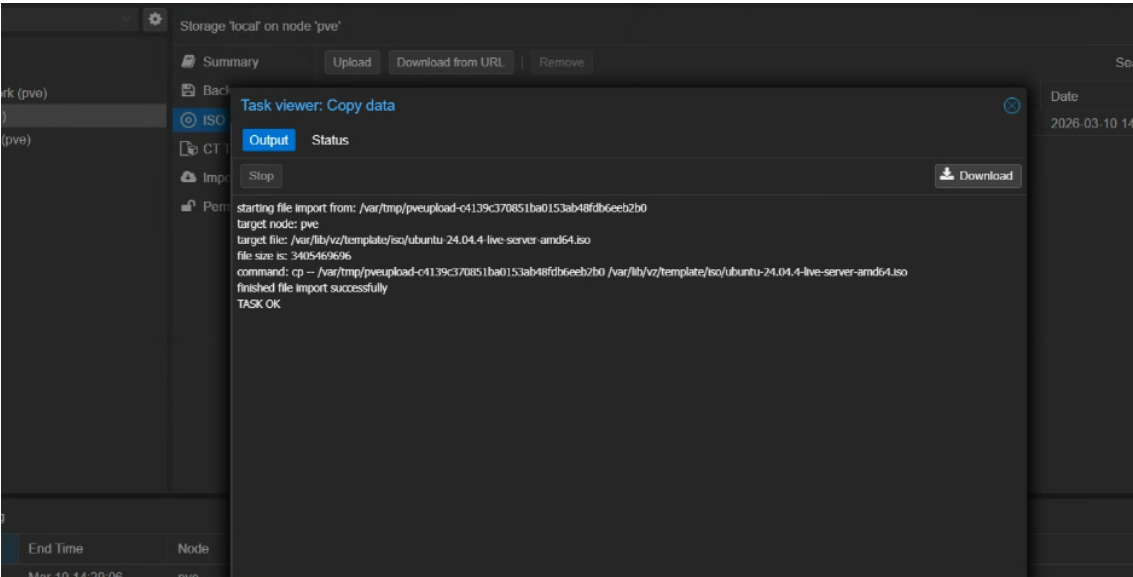

```
individual files in /usr/share/doc/*/copyright.
Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
root@pve:~# nano /etc/apt/sources.list
root@pve:~# cd /usr/share/javascript/proxmox-widget-toolkit
root@pve:/usr/share/javascript/proxmox-widget-toolkit# cp proxmoxlib.js proxmoxlib.js.bak
root@pve:/usr/share/javascript/proxmox-widget-toolkit# nano proxmoxlib.js
```

```
GNU nano 8.4 proxmoxlib.js *
    Ext.Msg.alert(gettext('Error'), response.htmlStatus);
},
success: function (response, opts) {
    let res = response.result;
    if (
        res === null ||
        res === undefined ||
        !res ||
        res.data.status.toLowerCase() !== 'active'
    ) {
        void [{
            title: gettext('No valid subscription'),
            icon: Ext.Msg.WARNING,
            message: Proxmox.Utils.getNoSubKeyHtml(res.data.url),
            buttons: Ext.Msg.OK,
            callback: function (btn) {
                if (btn !== 'ok') {
                    return;
                }
                orig_cmd();
            }
        }];
    }
}
```

Po zalogowaniu nie występuje okienko. 😊

Zad 2.

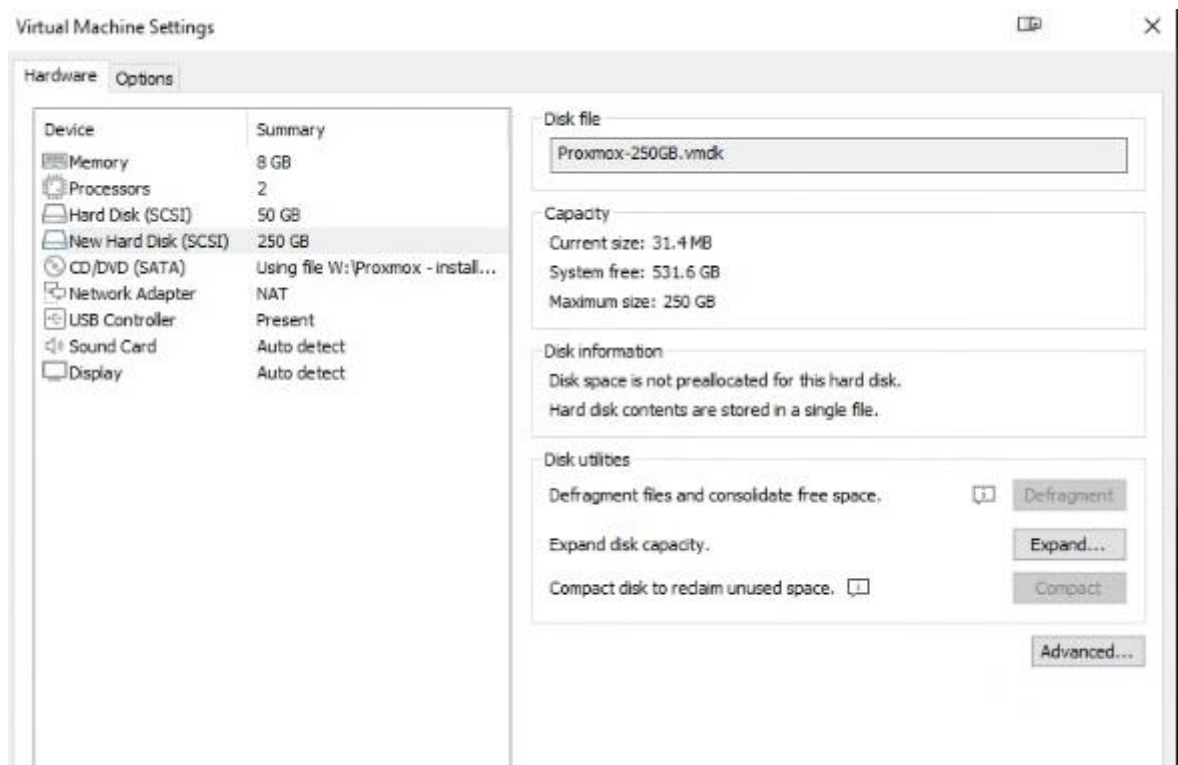
Import obrazu ISO systemu Ubuntu Server 24.04 oraz Windows Server do lokalnego magazynu Proxmox.



Summary	Upload	Download from URL	Remove	Search: Name, Format		
Backups						
ISO Images	ubuntu-24.04.4-live-server-amd64.iso	2026-03-10 14:29:06	iso	3.41 GB		
CT Templates	WinSrv_SERVER_EVAL_x64FRE_en-us.iso	2026-03-10 14:58:17	iso	5.65 GB		
Import						
Permissions						

Zad 3.

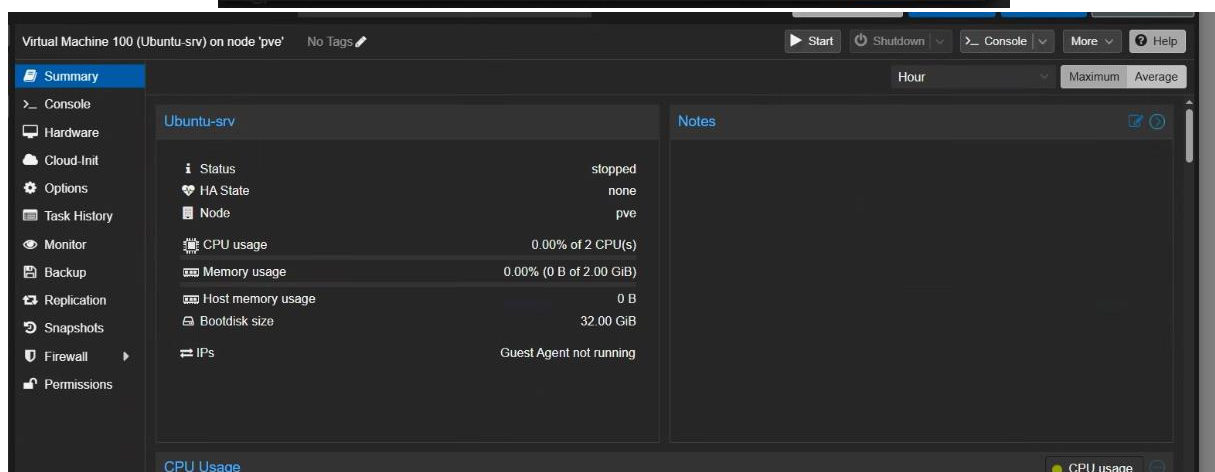
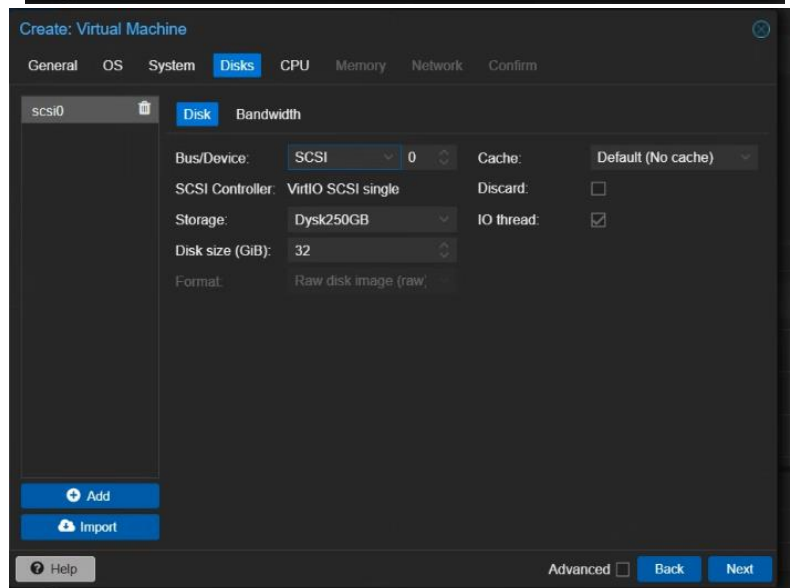
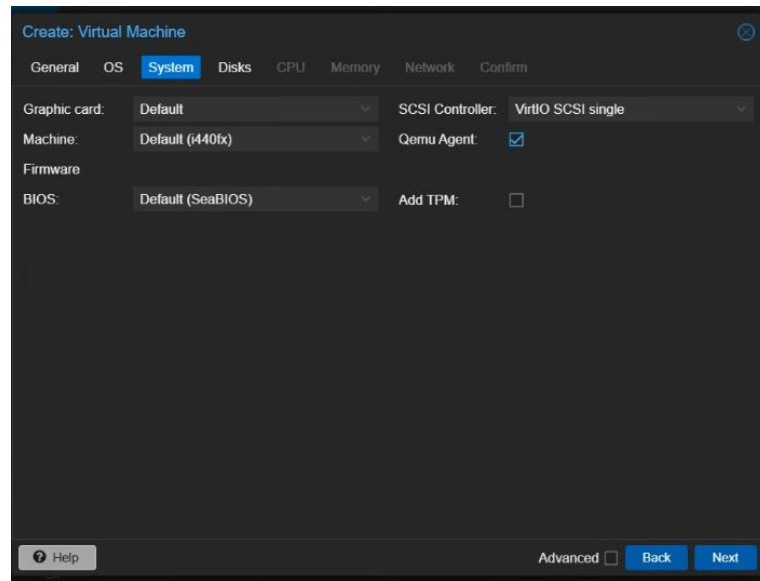
Dodanie nowego dysku o wielkości 250GB.



Reload		Create: Volume Group		No volume group selected			More	
Name		Number of LVs	Assigned to LVs	Size	Free			
✓ Disk250GB		1	100%	268.43 GB	130.02 MB			
/dev/sdb			100%	268.43 GB	130.02 MB			
✓ pve		2	100%	53.15 GB	0 B			
/dev/sda3			100%	53.15 GB	0 B			

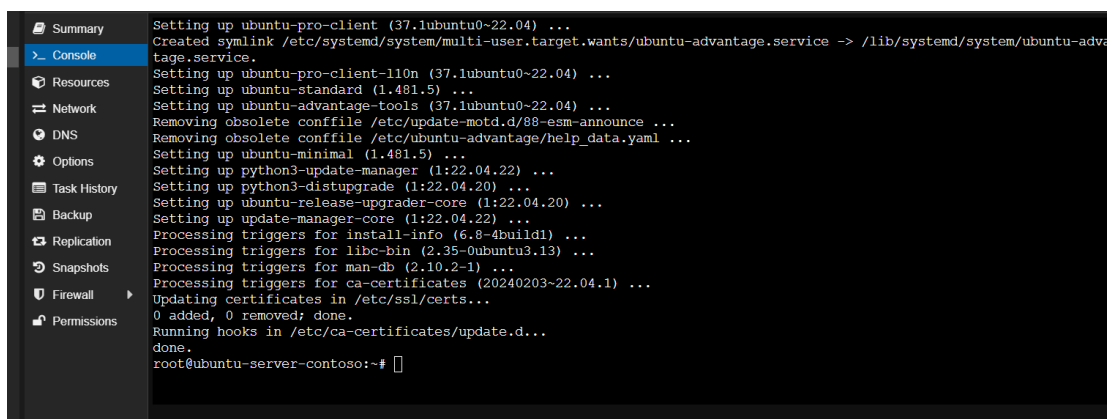
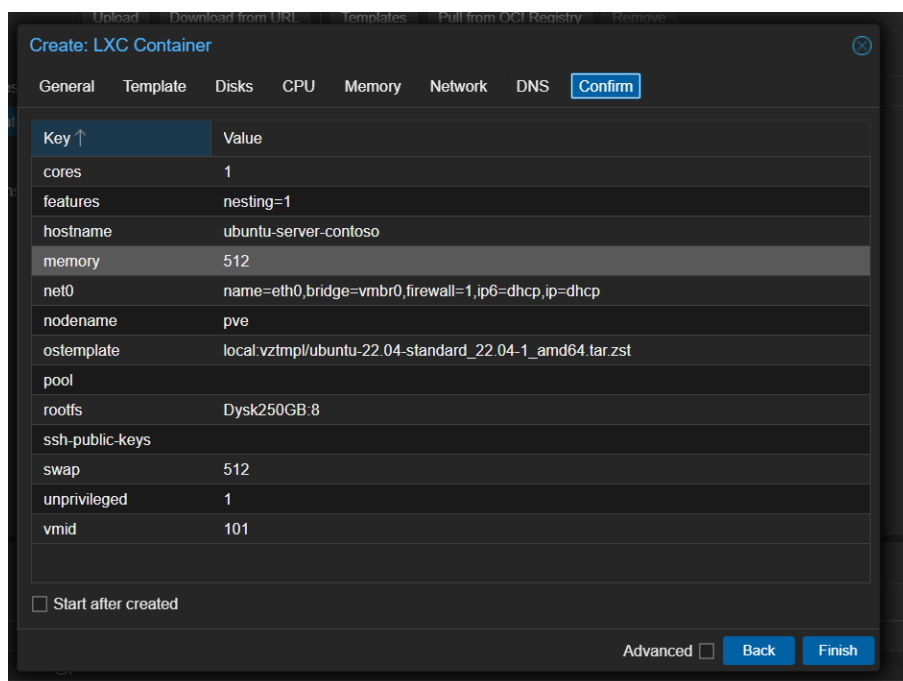
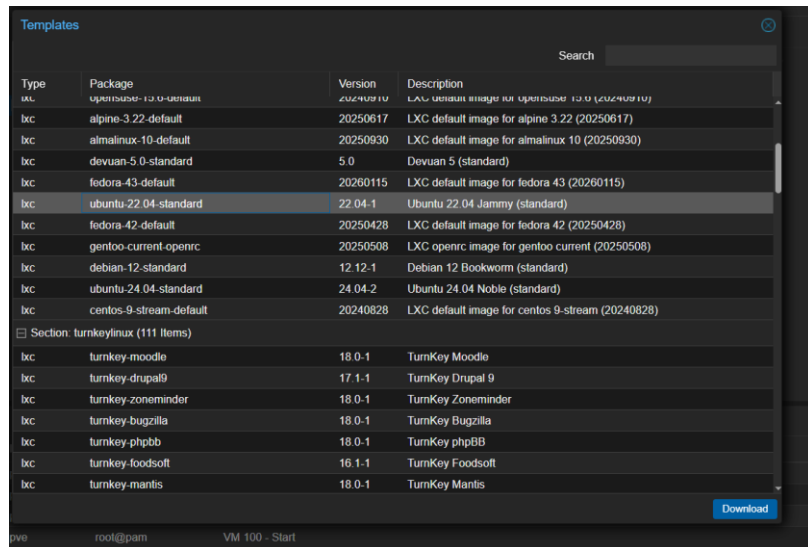
Zad 4.

Instalacja systemu Ubuntu Server.



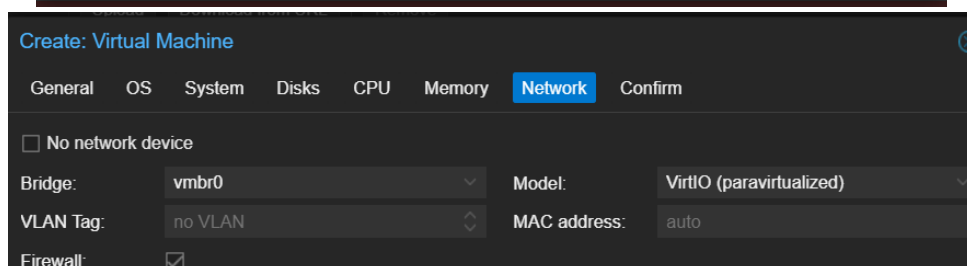
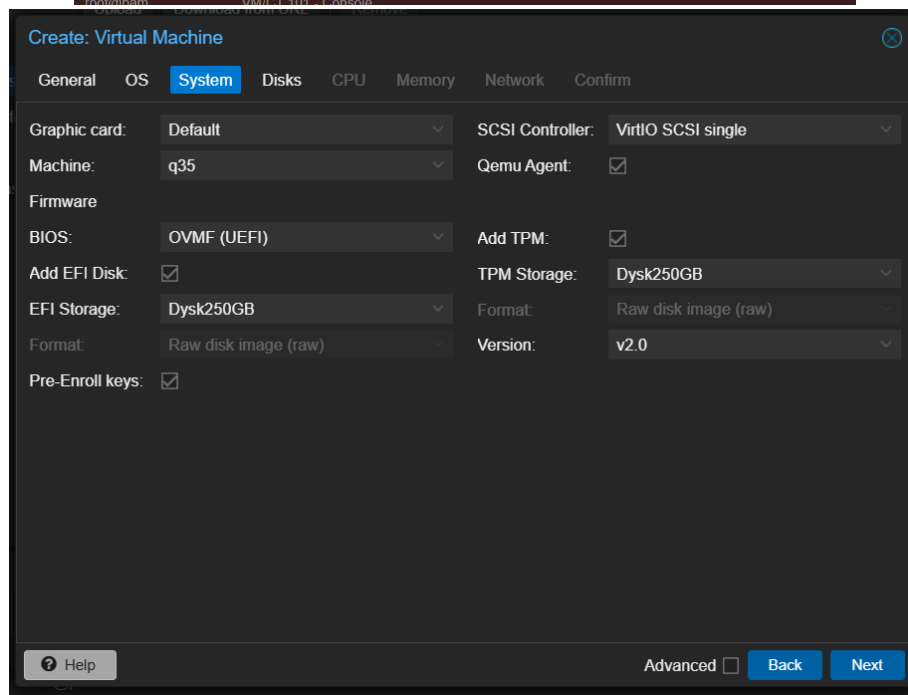
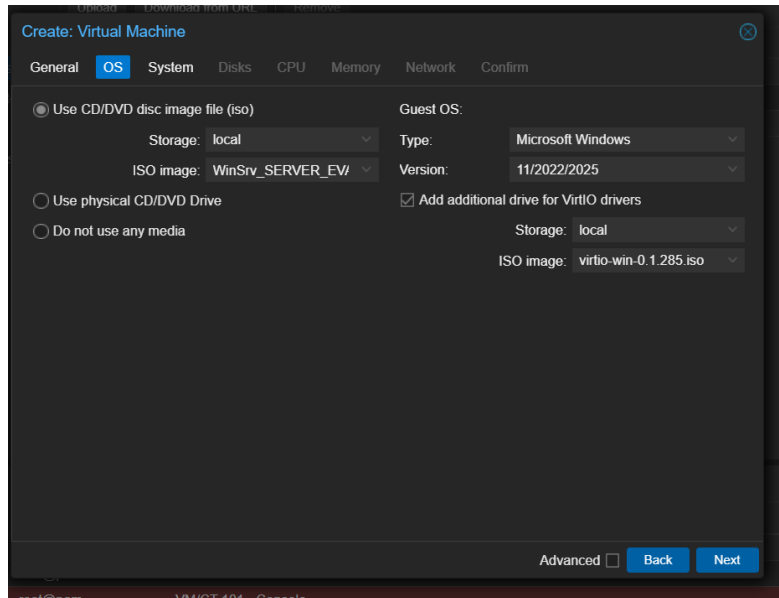
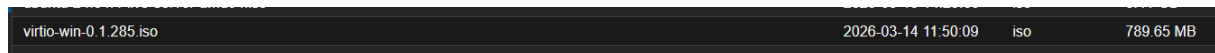
Zad 5.

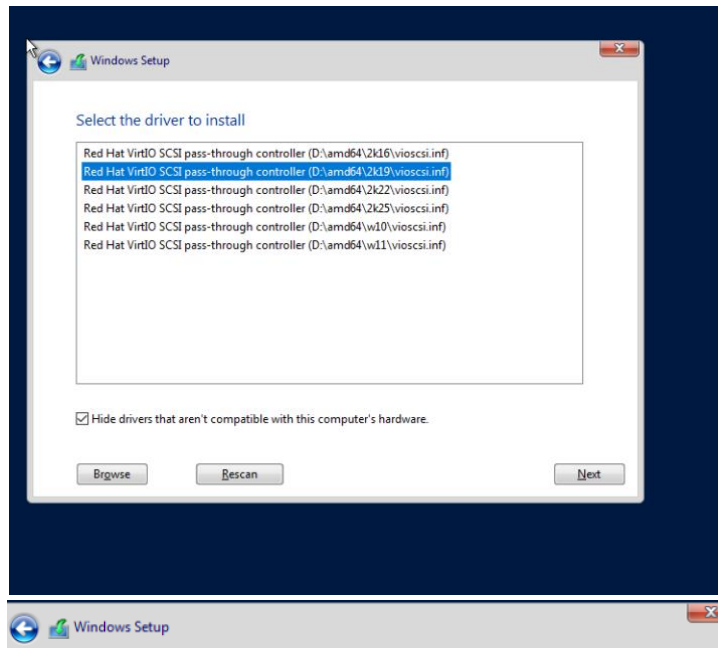
Tworzenie kontenera z systemem Ubuntu Server.



Zad 6.

Optymalizacja Windows (VirtIO). Instalacja sterowników parawirtualnych, aby dyski i sieć w Windows działały z natywną prędkością hypervisora.

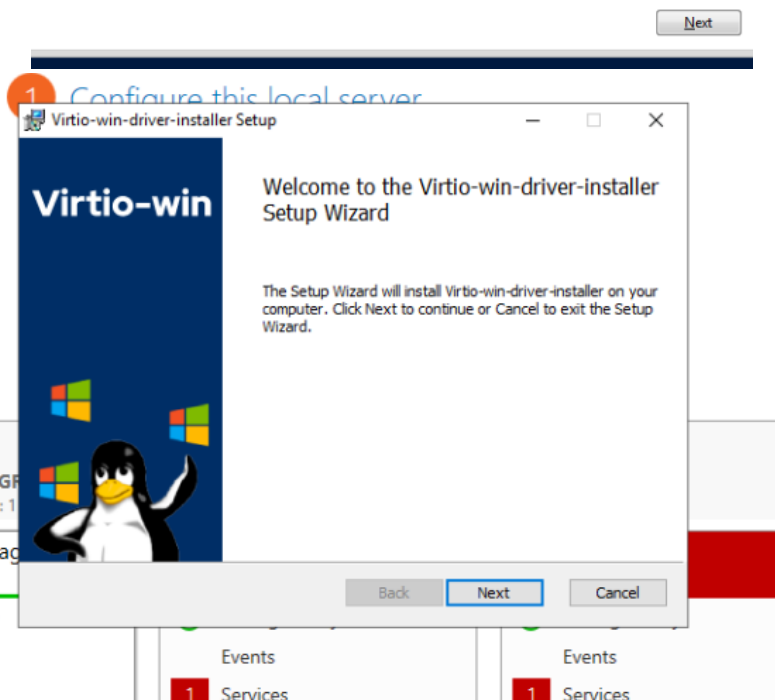




Where do you want to install Windows?

Name	Total size	Free space	Type
 Drive 0 Unallocated Space	100.0 GB	100.0 GB	

 Refresh  Delete  Format  New
 Load driver  Extend

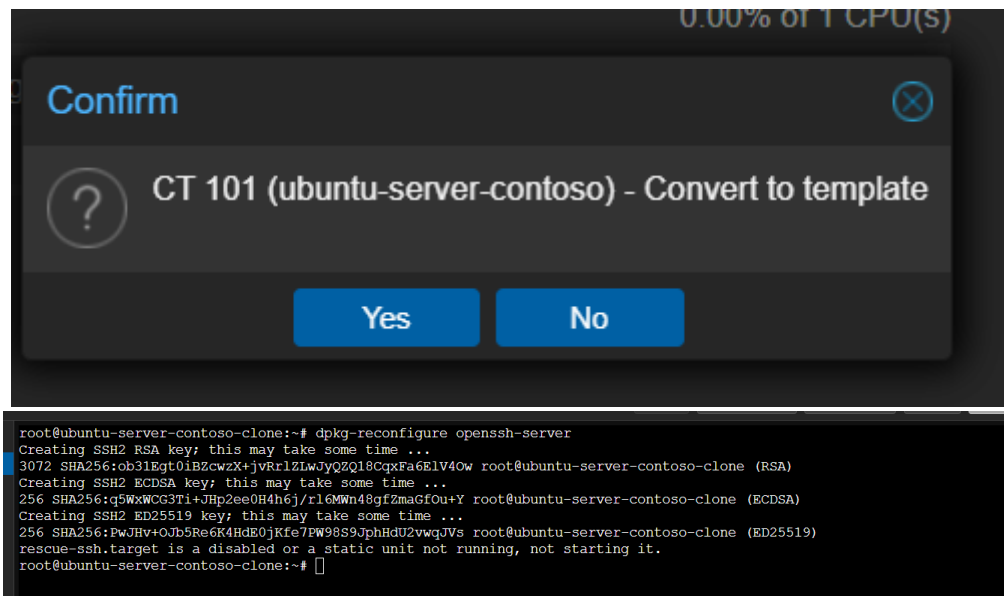


Automatyzacja Linux (Template & Cloud-Init). Przygotowanie "czystego" obrazu Ubuntu (bez unikalnych ID), który sam skonfiguruje użytkownika przy każdym nowym wdrożeniu.



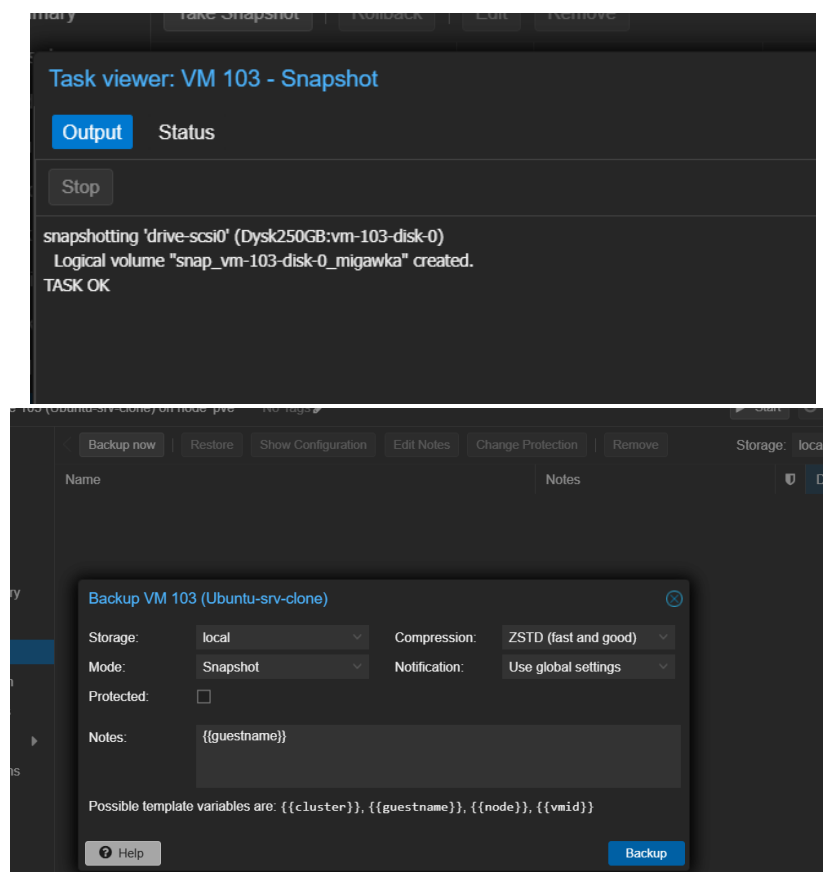
Zad 8.

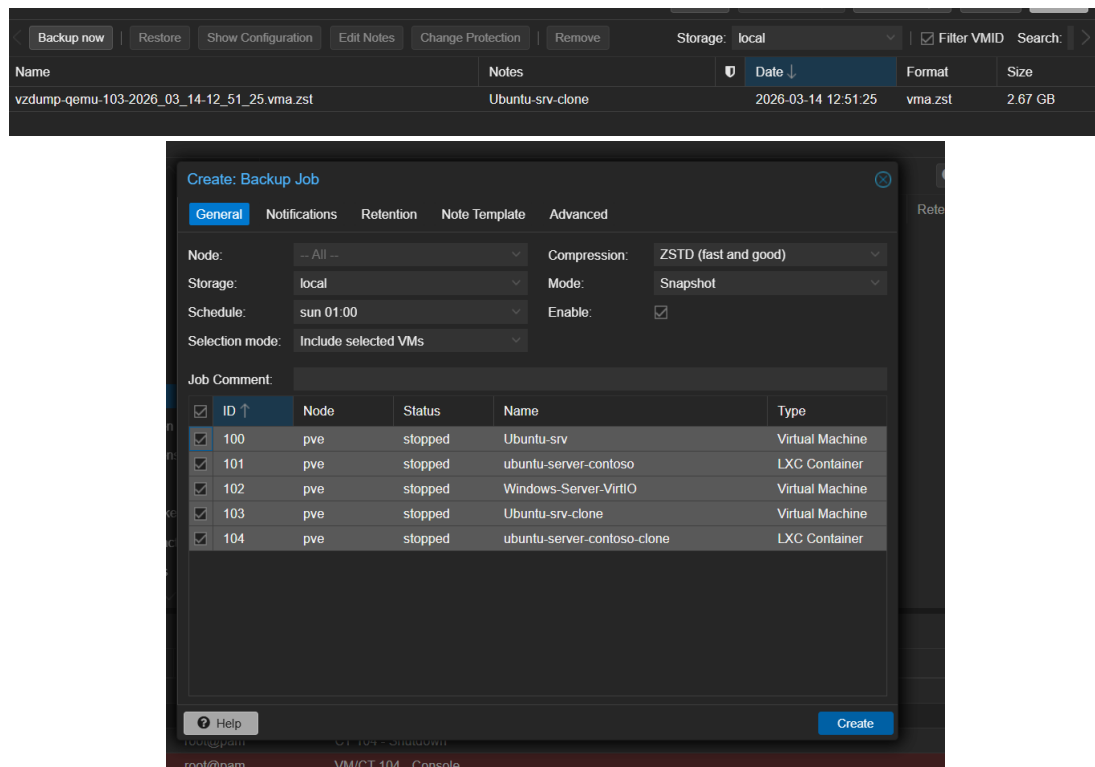
Szablony kontenerów (LXC). Szybkie klonowanie lekkich instancji systemowych i regeneracja kluczy bezpieczeństwa SSH po stronie klienta.



Zad 9.

Ciągłość biznesowa (Backup/Snapshot). Testowanie "punktów przywracania" przed zmianami oraz automatyzacja kopii zapasowych na wypadek awarii.





Zad 10.

Izolacja dostępu (PAM). Tworzenie użytkownika systemowego, który zarządza Proxmoxem przez WWW, ale nie ma wstępu do konsoli serwera (bezpieczeństwo powłoki).

```
root@pve:~# adduser jgula
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for jgula
Enter the new value, or press ENTER for the default
  Full Name []: Jacek
  Room Number []:
  Work Phone []:
  Home Phone []:
  Other []:
Is the information correct? [Y/n] Y

root@pve:~# usermod -s /bin/false jgula
```

Add: User

User name:

igula

First Name:

Jacek

Realm:

Linux PAM standard aut

Last Name:

Gula

Group:

E-Mail:

Expire:

never

Enabled:

☒

Comment:

Advanced

Add

Add: User Permission

Path:

/

User:

igula@pam

Role:

PVEVMUser

Propagate:

☒

Help

Add

Documentation

Create VM

Create CT

igula@pam

Reboot

Shutdown

Shell

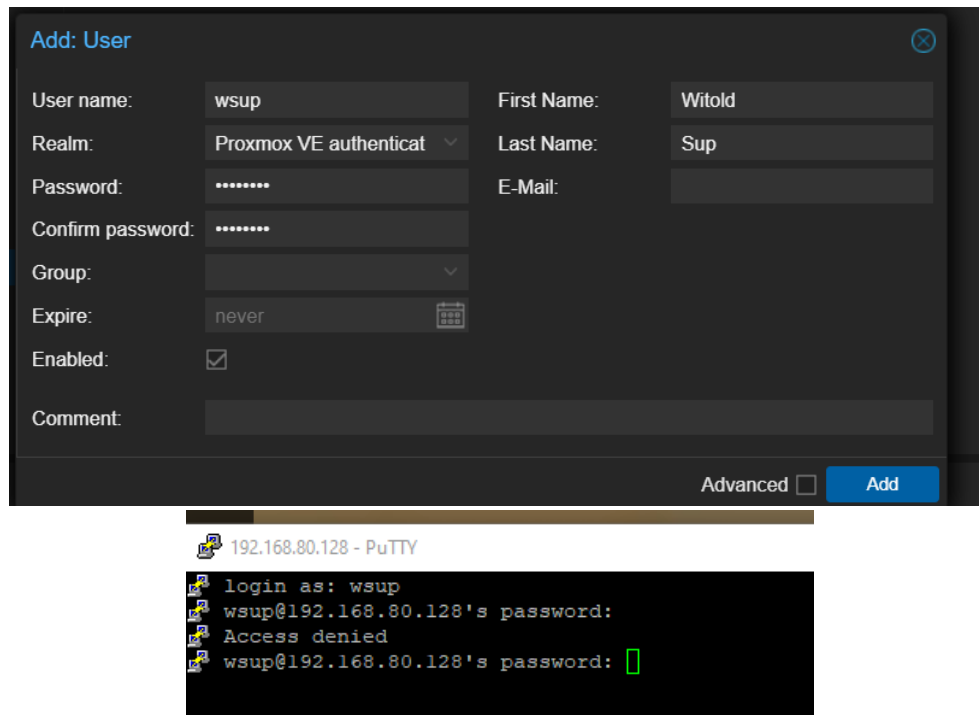
Bulk Actions

Help

Brak dostępu do konsoli

Zad 11.

Uwierzytelnianie PVE. Testowanie różnicy między kontem systemowym (PAM) a kontem tylko wewnątrz bazy Proxmoxa (PVE).



The image shows the 'Add: User' dialog box in Proxmox VE. The fields are filled as follows:

Field	Value
User name:	wsup
First Name:	Witold
Realm:	Proxmox VE authenticat
Last Name:	Sup
Password:	
E-Mail:	
Confirm password:	
Group:	
Expire:	never
Enabled:	☒
Comment:	

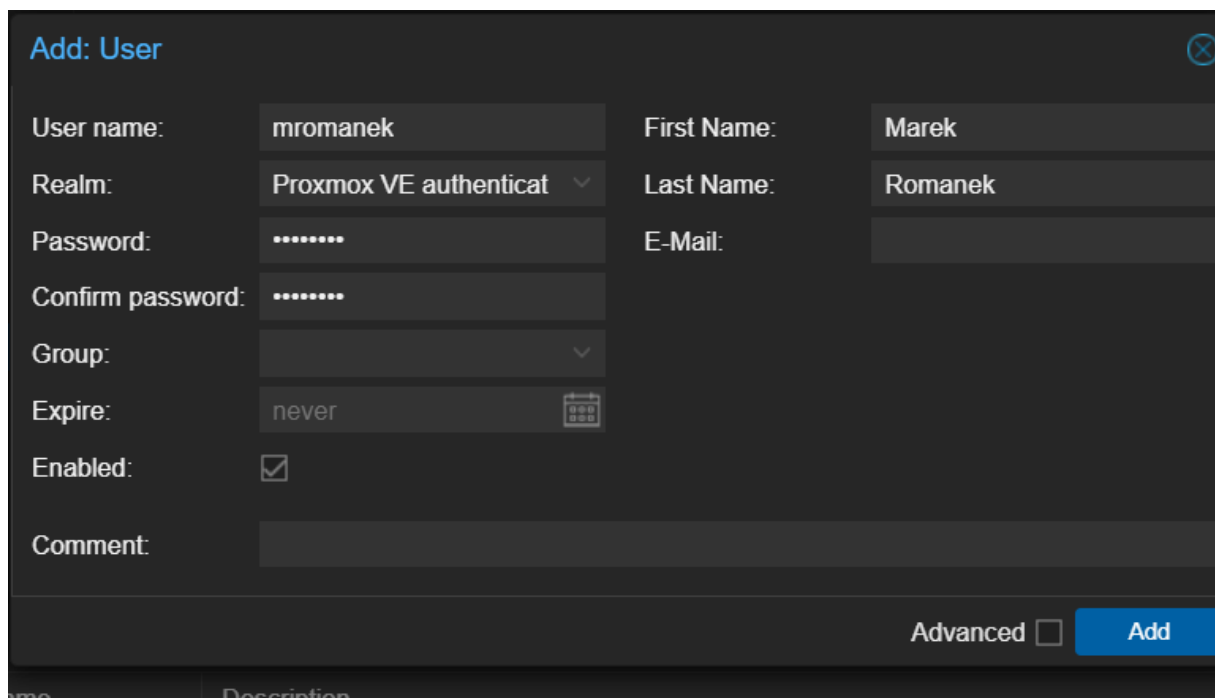
At the bottom right of the dialog are the 'Advanced' checkbox and the 'Add' button.

Below the dialog is a terminal window titled '192.168.80.128 - PuTTY' showing the following output:

```
login as: wsup
wsup@192.168.80.128's password:
Access denied
wsup@192.168.80.128's password: [ ]
```

Zad 12.

Testowanie uprawnień SSH. Weryfikacja, że tylko użytkownicy PAM mogą logować się bezpośrednio do systemu operacyjnego hosta.



The image shows the 'Add: User' dialog box in Proxmox VE. The fields are filled as follows:

Field	Value
User name:	mromanek
First Name:	Marek
Realm:	Proxmox VE authenticat
Last Name:	Romanek
Password:	
E-Mail:	
Confirm password:	
Group:	
Expire:	never
Enabled:	☒
Comment:	

At the bottom right of the dialog are the 'Advanced' checkbox and the 'Add' button.

Add: User Permission ✕

Path:

User:

Role:

Propagate: ☒

? Help Add

Documentation Create VM Create CT mromanek@pve

Reboot Shutdown Shell Bulk Actions ? Help

```
192.168.80.128 - PuTTY
login as: mromanek
mromanek@192.168.80.128's password:
Access denied
mromanek@192.168.80.128's password: 
```

Add: User ✕

User name: First Name:

Realm: Last Name:

Group:

Expire: 📅

Enabled: ☒

Comment:

Add: User Permission ✕

Path:

User:

Role:

Propagate: ☒

? Help Add

```
mromanek@pve: ~
login as: mromanek
mromanek@192.168.80.128's password:
Linux pve 6.17.13-1-pve #1 SMP PREEMPT_DYNAMIC PMX 6.17.13-1 (2026-02-10T14:06Z)
x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
mromanek@pve:~$ 
```

Zad 13.

Grupy i Rola PVEVMUser. Ograniczanie widoczności: użytkownik widzi i zarządza tylko konkretną maszyną (np. tylko Windows), a nie całym serwerem.

The image displays three screenshots of the Proxmox VE web interface, illustrating the configuration of a user and their permissions.

1. Create: Group

This dialog box is used to create a new group. The "Name" field is set to "UzytkownicyVM". The "Comment" field is empty. The "Create" button is visible at the bottom right.

2. Edit: User

This dialog box is used to edit a user. The "User name" is "wsup@pve". The "Group" is set to "UzytkownicyVM". The "First Name" is "Witold" and the "Last Name" is "Sup". The "Expire" field is set to "never". The "Enabled" checkbox is checked. The "Comment" field is empty. The "Advanced" checkbox is unchecked. The "OK" button is visible at the bottom right.

3. Add: Group Permission

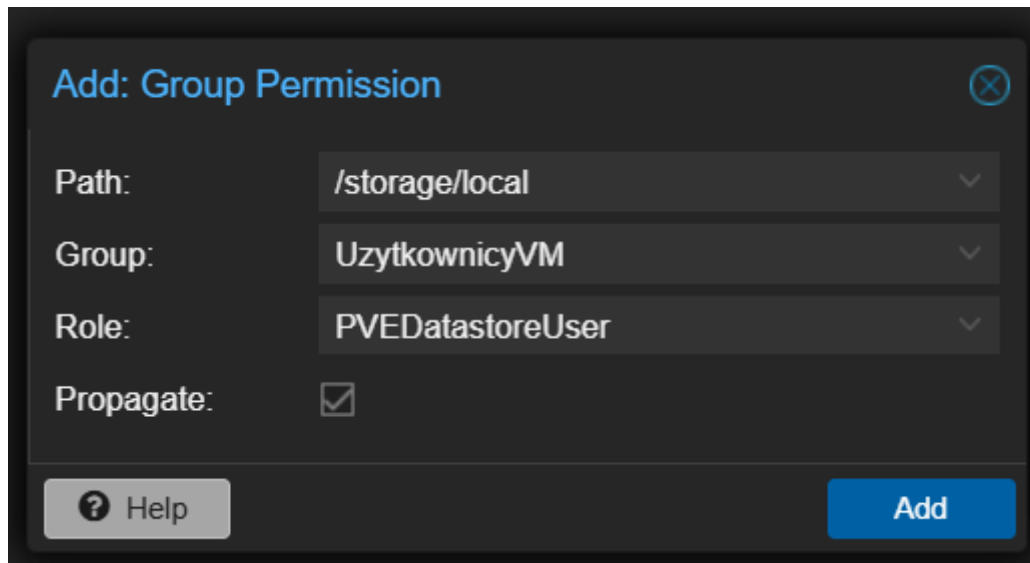
This dialog box is used to add a permission for a group. The "Path" is set to "/vms/102". The "Group" is set to "UzytkownicyVM". The "Role" is set to "PVEVMUser". The "Propagate" checkbox is checked. The "Add" button is visible at the bottom right.

4. Server View

This screenshot shows the "Server View" of the Proxmox VE interface. The "Datacenter" is expanded, showing the "pve" host. The "102 (Windows-Server-VirtIO)" virtual machine is listed under the "pve" host.

Zad 14.

Dostęp do zasobów (Datastore). Nadawanie uprawnień do "magazynu", aby użytkownicy mogli samodzielnie montować obrazy ISO (płyty instalacyjne).



Add: Group Permission

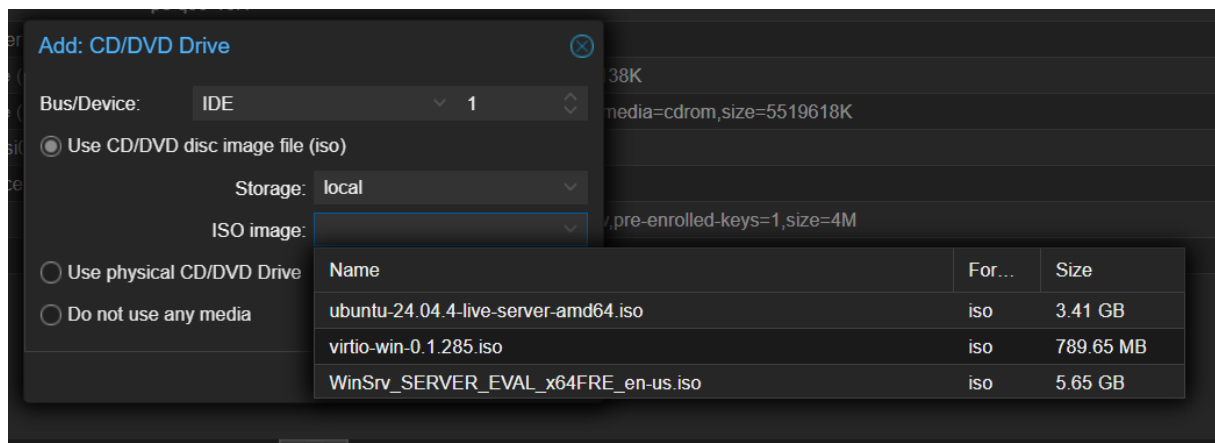
Path:

Group:

Role:

Propagate: ☒

[? Help](#) [Add](#)



Add: CD/DVD Drive

Bus/Device:

☒ Use CD/DVD disc image file (iso)

Storage:

ISO image:

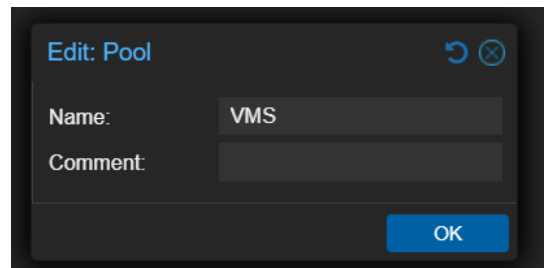
☐ Use physical CD/DVD Drive

☐ Do not use any media

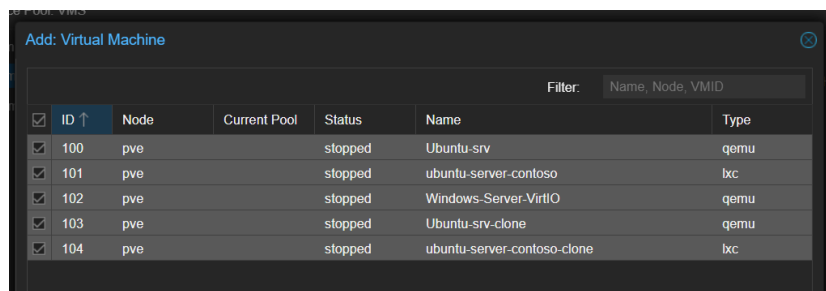
Name	For...	Size
ubuntu-24.04.4-live-server-amd64.iso	iso	3.41 GB
virtio-win-0.1.285.iso	iso	789.65 MB
WinSrv_SERVER_EVAL_x64FRE_en-us.iso	iso	5.65 GB

Zad 15.

Pule zasobów (Pools). Logiczne grupowanie maszyn w "worki", co pozwala nadać uprawnienia do wielu maszyn naraz jednym kliknięciem.

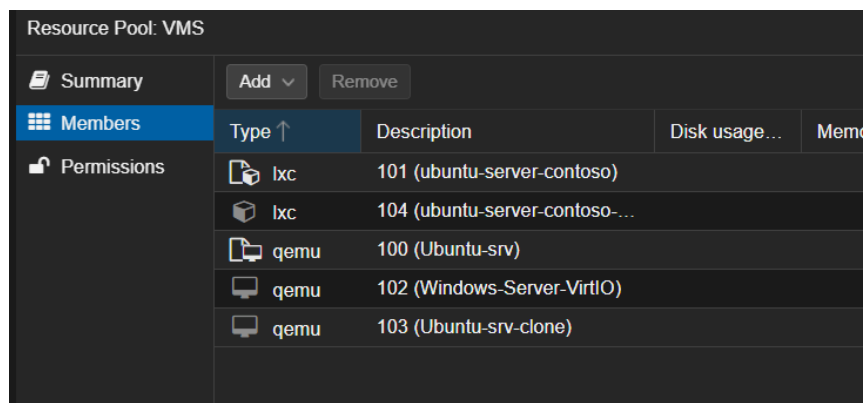


Dialog box titled "Edit: Pool" with a refresh icon and a close icon. It contains two input fields: "Name:" with the value "VMS" and "Comment:" which is empty. An "OK" button is at the bottom right.



Dialog box titled "Add: Virtual Machine" with a close icon. It features a filter input field containing "Name, Node, VMID". Below is a table with columns: ID, Node, Current Pool, Status, Name, and Type. Five rows are listed, all with status "stopped".

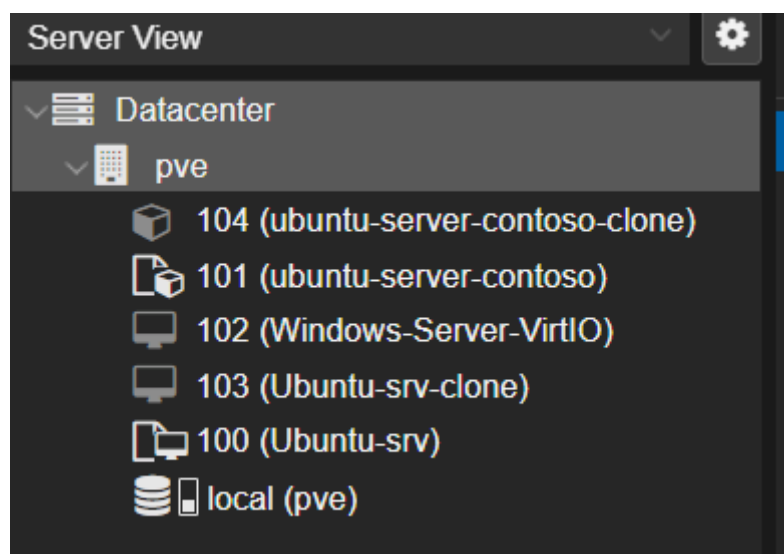
☒	ID ↑	Node	Current Pool	Status	Name	Type
☒	100	pve		stopped	Ubuntu-srv	qemu
☒	101	pve		stopped	ubuntu-server-contoso	lxc
☒	102	pve		stopped	Windows-Server-VirtIO	qemu
☒	103	pve		stopped	Ubuntu-srv-clone	qemu
☒	104	pve		stopped	ubuntu-server-contoso-clone	lxc



Resource Pool: VMS. The "Members" tab is selected. It shows a table of VMs with columns: Type, Description, Disk usage, and Memory. The table lists five VMs with their respective types and descriptions.

Type ↑	Description	Disk usage...	Mem...
lxc	101 (ubuntu-server-contoso)		
lxc	104 (ubuntu-server-contoso-...)		
qemu	100 (Ubuntu-srv)		
qemu	102 (Windows-Server-VirtIO)		
qemu	103 (Ubuntu-srv-clone)		

/pool/VMS @UzytkownicyVM PVEVMUser

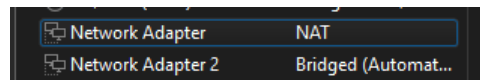


Server View sidebar showing a tree structure. "Datacenter" is expanded, showing "pve". Under "pve", there is a list of VMs with their IDs and names, each preceded by a small icon representing its type (lxc or qemu). At the bottom is "local (pve)" with a storage icon.

Datacenter	
pve	
104 (ubuntu-server-contoso-clone)	
101 (ubuntu-server-contoso)	
102 (Windows-Server-VirtIO)	
103 (Ubuntu-srv-clone)	
100 (Ubuntu-srv)	
local (pve)	

Zad 16.

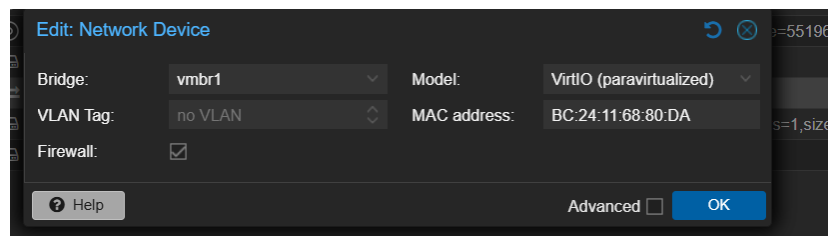
Passthrough/Dedicated NIC. Przypisanie fizycznej karty sieciowej wyłącznie dla jednej maszyny wirtualnej (bezpośredni dostęp do LAN).



```
root@pve:~# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: nic0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel master vmbr0 state UP group default qlen 1000
    link/ether 00:0c:29:3f:14:a4 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altname enx00c293f14a4
3: ens37: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN group default qlen 1000
    link/ether 00:0c:29:3f:14:ae brd ff:ff:ff:ff:ff:ff
    altname enp2s5
    altname enx00c293f14ae
4: vmbr0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default qlen 1000
    link/ether 00:0c:29:3f:14:a4 brd ff:ff:ff:ff:ff:ff
    inet 192.168.80.128/24 scope global vmbr0
        valid_lft forever preferred_lft forever
    inet6 fe80::20c:29ff:fe3f:14a4/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
root@pve:~#
```

```
auto vmbr1
iface vmbr1 inet dhcp
    bridge-ports ens34
    bridge-stp off
    bridge-fd 0
```

```
root@pve:~# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: nic0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel master vmbr0 state UP group default qlen 1000
    link/ether 00:0c:29:3f:14:a4 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altname enx00c293f14a4
3: ens37: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN group default qlen 1000
    link/ether 00:0c:29:3f:14:ae brd ff:ff:ff:ff:ff:ff
    altname enp2s5
    altname enx00c293f14ae
5: vmbr0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default qlen 1000
    link/ether 00:0c:29:3f:14:a4 brd ff:ff:ff:ff:ff:ff
    inet 192.168.80.128/24 scope global vmbr0
        valid_lft forever preferred_lft forever
    inet6 fe80::20c:29ff:fe3f:14a4/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
6: vmbr1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UNKNOWN group default qlen 1000
    link/ether 86:1a:be:ff:d5:04 brd ff:ff:ff:ff:ff:ff
    inet6 fe80::841a:beff:feff:d504/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
root@pve:~#
```



Zad 17.

Izolacja (Internal Network). Tworzenie wirtualnego switcha bez wyjścia na świat, aby maszyny rozmawiały tylko ze sobą (bezpieczny poligon).

```
auto vmbr2
iface vmbr2 inet static
    address 172.16.0.1/24
    bridge-ports none
    bridge-stp off
    bridge-fd 0
```

Edit: Network Device

Bridge:	vmbr2	Model:	VirtIO (paravirtualized)
VLAN Tag:	no VLAN	MAC address:	BC:24:11:68:80:DA
Firewall:	☒		

☐ Advanced

Add: Network Device (veth)

Name:	Linux-Bridge	IPv4:	☒ Static ☐ DHCP
MAC address:	auto	IPv4/CIDR:	None
Bridge:	vmbr2	Gateway (IPv4):	
VLAN Tag:	no VLAN	IPv6:	☒ Static ☐ DHCP ☐ SLAAC
Firewall:	☒	IPv6/CIDR:	None
		Gateway (IPv6):	

☐ Advanced

```
root@ubuntu-server-contoso-clone:~# ping 172.16.0.20
PING 172.16.0.20 (172.16.0.20) 56(84) bytes of data.
64 bytes from 172.16.0.20: icmp_seq=1 ttl=64 time=0.015 ms
64 bytes from 172.16.0.20: icmp_seq=2 ttl=64 time=0.036 ms
64 bytes from 172.16.0.20: icmp_seq=3 ttl=64 time=0.036 ms
64 bytes from 172.16.0.20: icmp_seq=4 ttl=64 time=0.036 ms
```

Zad 18.

Maskarada (NAT). Konfiguracja routera wewnątrz Proxmoxa, aby maszyny z prywatnej podsieci mogły pobierać dane z internetu.

```
auto vmbr3
iface vmbr3 inet static
    address 10.10.10.1/24
    bridge-ports none
    bridge-stp off
    bridge-fd 0

    post-up echo 1 > /proc/sys/net/ipv4/ip_forward
    post-up iptables -t nat -A POSTROUTING -s '10.10.10.0/24' -o eno1 -j MASQUERADE
    post-down iptables -t nat -D POSTROUTING -s '10.10.10.0/24' -o eno1 -j MASQUERADE
```



Network Device (net0)

virtio=BC:24:11:68:80:DA,bridge=vmbr3

Internet Protocol Version 4 (TCP/IPv4) Properties

General

You can get IP settings assigned automatically if your network supports this capability. Otherwise, you need to ask your network administrator for the appropriate IP settings.

☐ Obtain an IP address automatically

☒ Use the following IP address:

IP address: 10 . 10 . 10 . 10
Subnet mask: 255 . 255 . 255 . 0
Default gateway: 10 . 10 . 10 . 1

☐ Obtain DNS server address automatically

☒ Use the following DNS server addresses:

Preferred DNS server: 8 . 8 . 8 . 8

```
C:\Users\Administrator>ping 10.10.10.1
```

```
Pinging 10.10.10.1 with 32 bytes of data:
```

```
Reply from 10.10.10.1: bytes=32 time<1ms TTL=64
```

```
Reply from 10.10.10.1: bytes=32 time<1ms TTL=64
```

```
Reply from 10.10.10.1: bytes=32 time<1ms TTL=64
```

```
Reply from 10.10.10.1: bytes=32 time<1ms TTL=64
```

```
Ping statistics for 10.10.10.1:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
```

```
    Approximate round trip times in milli-seconds:
```

```
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Zad 19.

Niezawodność (Bonding). Łączenie dwóch kart sieciowych w jedną (Active-Backup) – jeśli jedna się spali, sieć działa dalej bez przerwy.

```
Network Adapter 3    NAT
Network Adapter 4    NAT

auto bond0
iface bond0 inet dhcp
    slaves ens38 ens39
    bond-mode active-backup
    bond-miimon 100
    bond-primary ens38
```

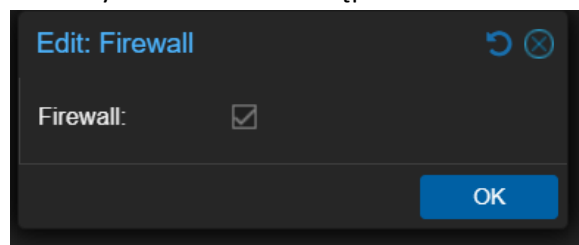
Zad 20.

Segmentacja (VLAN). Wpisanie maszyny wirtualnej w konkretny identyfikator VLAN (100), aby odseparować ruch sieciowy na poziomie warstwy 2.

```
auto vmbr2.100
iface vmbr2.100 inet dhcp
    vlan-raw-device ens40
```

Zad 21.

Ochrona Host'a. Aktywacja zapory na poziomie całego klastra i odblokowanie protokołu ICMP (Ping), aby monitorować dostępność serwera.



```
permitted by applicable law.
root@pve:~# cat /etc/pve/firewall/cluster.fw
[OPTIONS]

enable: 1
```

```
root@pve:~# iptables -Lvn
iptables: No chain/target/match by that name.
root@pve:~#
```

```
C:\Users\Admin>ping 192.168.80.128

Pinging 192.168.80.128 with 32 bytes of data:
Request timed out.
```

Add: Rule

Direction:
in

Enable:
☒

Action:
ACCEPT

Macro:

Interface:

Protocol:
icmp

Source:

Source port:

Destination:

ICMP type:

Comment:
Allow ping

Advanced ☐
Add

```

C:\Users\Admin>ping 192.168.80.128

Pinging 192.168.80.128 with 32 bytes of data:
Reply from 192.168.80.128: bytes=32 time<1ms TTL=64
Reply from 192.168.80.128: bytes=32 time<1ms TTL=64

```

Zad 22.

Aliasy i IPSet. Grupowanie adresów IP w listy (IPSet), co pozwala jedną regułą blokować lub wpuszczać całe działy firmy.

Name	IP/CIDR ↑	Comment
myPC	192.168.80.1/24	myPC

Add: Rule

Direction:
in

Enable:
☒

Action:
DROP

Macro:

Interface:

Protocol:
icmp

Source:
mypc

Source port:

Destination:

ICMP type:

Comment:

Advanced ☐
Add

```
C:\Users\Admin>ping 192.168.80.128

Pinging 192.168.80.128 with 32 bytes of data:
Request timed out.
```

IPSet: Create Remove Edit		IP/CIDR: Add Remove Edit	
IPSet ↑	Comment		
myips		1	dc/ip2
		2	dc/ip3
		3	dc/myipc

Zad 23.

Ochrona Maszyn (VM Firewall). Precyzyjne otwieranie tylko niezbędnych portów (np. port 22 dla SSH) dla konkretnej wirtualnej maszyny.

```
192.168.80.128 - PuTTY
login as: root
root@192.168.80.128's password:
Linux pve 6.17.13-1-pve #1 SMP PREEMPT_DYNAMIC PMX 6.17.13-1 (2026-02-10T14:06Z)
x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Sat Mar 14 14:56:30 2026 from 192.168.80.1
root@pve:~#
```